



THE 3-GATE MORPHOLOGICAL REACHABILITY & SINGULARITY VALIDATOR

Screening Untrusted Stochastic VLM Proposals Before Committing Downstream Motion Trajectories



UNTRUSTED PROPOSAL

VLM Semantic Intent

Pose: $T_{\text{target}}^{\text{world}} \in SE(3)$

Coord: $p_{\text{target}} \in \mathbb{R}^3$

Aperture: d_{grasp}

Confidence: $p_{\text{sem}} \geq 0.75$

Stochastic prediction from 2D pixels



GATE 1

Workspace Envelope

Geometric Feasibility

Reach: $\|p_{\text{target}} - p_{\text{base}}\| \leq R_{\text{max}}$

Floor: $z_{\text{target}} \geq z_{\text{table_min}}$

Joint Bounds: $q_{\text{min}} \leq q \leq q_{\text{max}}$

IK Solver: Analytical / Damped LS

Rejects points outside physical arm reach



GATE 2

Singularity Filter

Jacobian Conditioning

Condition: $\kappa(J) = \frac{\sigma_{\text{max}}}{\sigma_{\text{min}}} \leq 50.0$

Manipulability: $w = \sqrt{\det(JJ^T)} \geq w_{\text{min}}$

Joint Velocity: $\|\dot{q}\| = \|J^{-1}\dot{x}\| \leq \dot{q}_{\text{max}}$

Safety Margin: Non-zero nullspace

Prevents infinite joint velocity spikes



GATE 3

Static Torque Limits

Actuator Thermal Safety

Gravity Load: $\tau_{\text{grav}}(q) \leq \tau_{\text{cont}}$

Payload Wrench: $J(q)^T F_{\text{load}} \leq \tau_{\text{margin}}$

Thermal Rise: $T_{\text{motor}} < 85^\circ\text{C}$

Current Draw: $I_{\text{phase}} \leq I_{\text{continuous}}$

Guards motor windings against stall burn



ALL GATES PASSED: ISSUE INTENT LEASE ($\mathcal{L}_{\text{intent}}$)

Commit validated 3D target frame and workspace envelope to System 1.5 Trajectory Planner



GATE FAILED: EMIT STRUCTURED ABSTENTION

Revoke intent proposal; command Category 2 Safe Position Hold; prompt operator for clarification